

NETWORKING FUNDAMENTALS II





COURSE STRUCTURE

Day 1: Networking Fundamentals I

Day 2: Networking Fundamentals II

Day 3: Networking Fundamentals II

Day 4: Network Security in Depth

Day 5: Routing Fundamentals & Capstone

AGENDA FOR DAY 2

1. DHCP: Automated IP Assignment
2. DNS: Domain Name System Deep Dive
3. NAT: Network Address Translation
4. Network Security: ACLs, Firewalls & Port Security
5. Wireless Fundamentals & Wi-Fi Security
6. VPN Basics



DHCP, NAT, DNS

DHCP CONCEPTS

- Dynamic Host Configuration Protocol (DHCP):
- Automates IP address assignment
- Reduces manual configuration errors
- Provides additional configuration: subnet mask, gateway, DNS, etc.
- Operates on UDP ports 67 (server) and 68 (client)

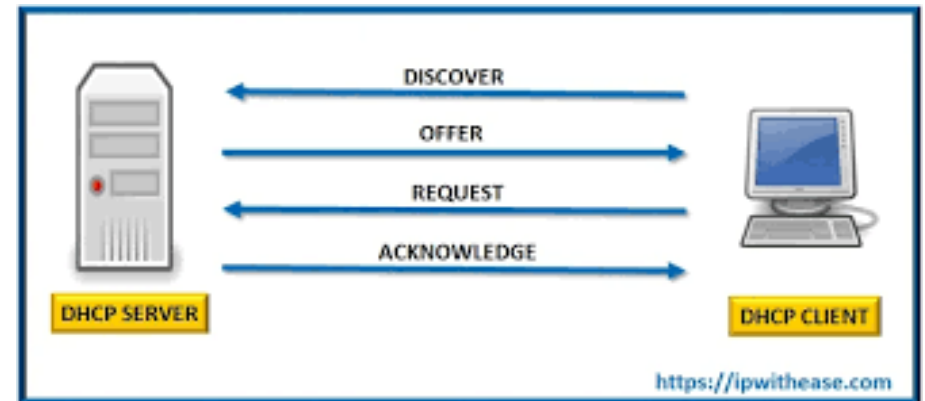


DHCP SCOPES AND EXCLUSIONS

- DHCP Scope:
 - Range of IP addresses that a DHCP server can assign
 - Defines the subnet, default gateway, lease duration, and DNS servers
- Exclusions:
 - IP addresses not assigned automatically by DHCP
 - Used for devices that need static IPs (e.g., servers, printers, routers)

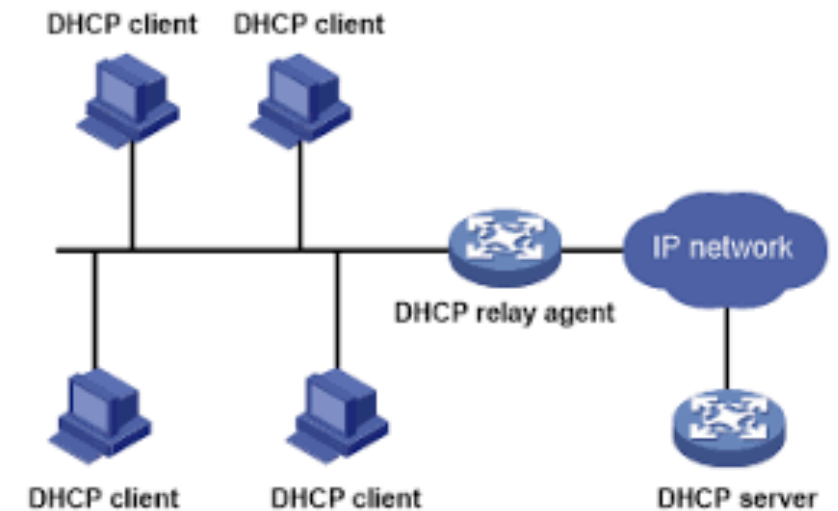
DHCP LEASE PROCESS (DORA)

- Discover – Client broadcasts to find a DHCP server.
- Offer – Server offers an available IP address.
- Request – Client requests the offered IP.
- Acknowledge – Server confirms and finalizes the lease.

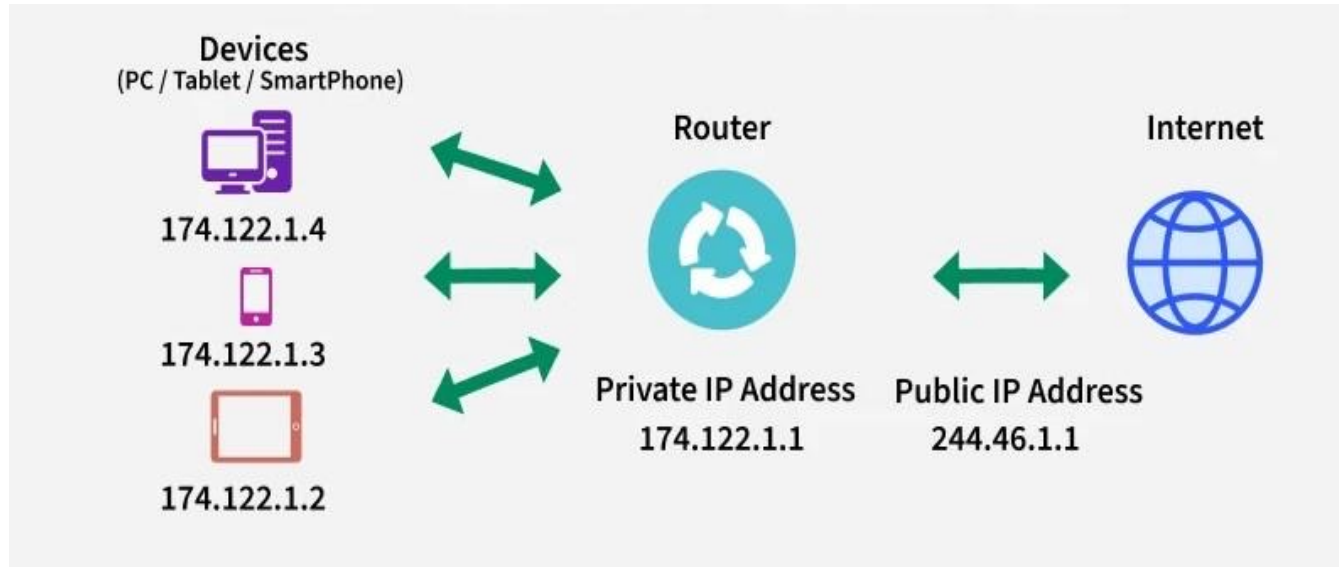


DHCP RELAY (IP HELPER-ADDRESS)

- Purpose:
 - DHCP messages are broadcasts — routers don't forward them by default.
 - DHCP Relay Agent (configured using ip helper-address) forwards requests to the DHCP server.



NAT OVERVIEW



- Network Address Translation
- Translates private IP addresses to public IPs
- Allows multiple internal devices to share limited public IP space
- Improves security by hiding internal network details

NAT TYPES

- Static NAT: One-to-one mapping between internal and external Ips
- Permanent translation manually configured
- Example:
 - `ip nat inside source static 192.168.10.10 203.0.113.10`
- Dynamic NAT: Many-to-many mapping using a pool of public IPs
- IPs are assigned from a pool when sessions are initiated
- Example:
 - `ip nat pool PUBLIC_POOL 203.0.113.10 203.0.113.20 netmask 255.255.255.0`
 - `ip nat inside source list 1 pool PUBLIC_POOL`
 - `access-list 1 permit 192.168.10.0 0.0.0.255`

NAT TYPES

- PAT (a.k.a. NAT Overload):
 - Many-to-one translation using port numbers
 - Multiple private IPs share a single public IP
 - Example:
 - ip nat inside source list 1 interface GigabitEthernet0/0/2

```
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/2
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#access-list 1 permit 10.0.0.4
Router(config)#access-list 1 permit 10.0.0.5
Router(config)#access-list 1 permit 10.0.0.6
Router(config)#ip nat inside source list 1 interface GigabitEthernet0/0/2
```

INTRODUCTION TO DNS

- DNS translates human-readable names into IP addresses.
- Hierarchical structure: Root → TLD → Domain → Subdomain.
- Enables scalable and distributed resolution.
- Used both publicly (internet) and privately (VPC internal DNS).



www.google.com.

www → subdomain

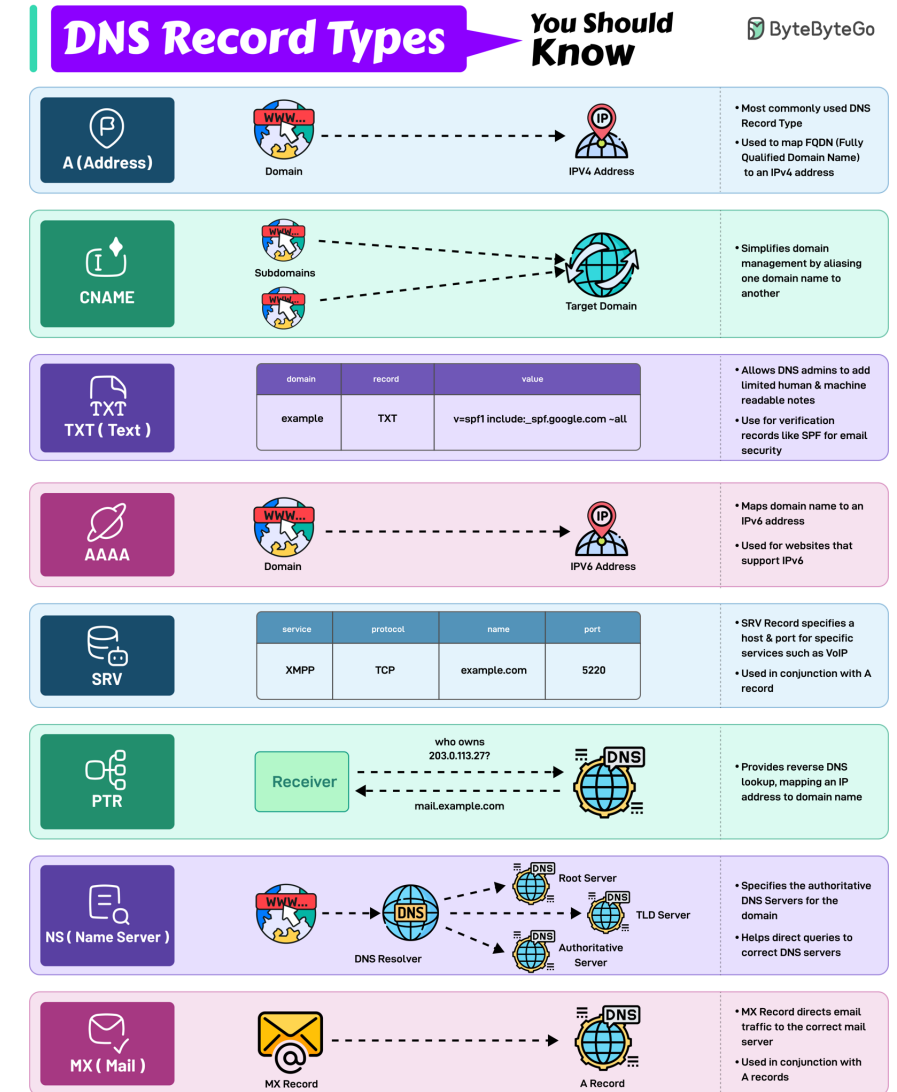
google → domain

com → TLD

. → root (the top of the DNS hierarchy)

DNS RECORD TYPES

- A / AAAA: Maps name to IPv4/IPv6 address.
- CNAME: Alias to another DNS name.
- MX: Mail exchange records.
- TXT: Text data (e.g., SPF, DKIM).
- Alias: AWS-specific pointer to load balancers, CloudFront, etc.



POP QUIZ:

Which UDP ports does DHCP use for communication?

- A. 80 and 443
- B. 67 and 68
- C. 110 and 143
- D. 22 and 23



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POP QUIZ:

What is the correct order of the DHCP lease process?

- A. Offer, Request, Acknowledge, Discover
- B. Discover, Offer, Request, Acknowledge
- C. Discover, Request, Offer, Acknowledge
- D. Request, Offer, Discover, Acknowledge



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POP QUIZ:

What type of NAT allows multiple private IPs to share one public IP using port numbers?

- A. Static NAT
- B. Dynamic NAT
- C. PAT (NAT Overload)
- D. Reverse NAT



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NETWORK SECURITY

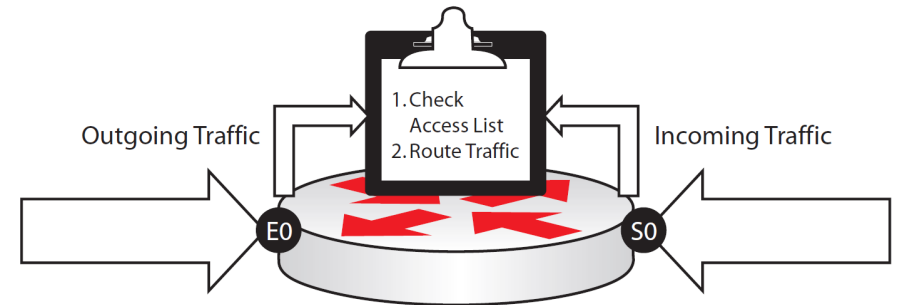
THE CIA TRIAD & DEFENSE (C.I.A)

- **Confidentiality** — data accessible only to authorized users (encryption, access control)
- **Integrity** — data cannot be modified without detection (hashing: SHA-256, digital signatures)
- **Availability** — systems accessible when needed (redundancy, backups, DDoS protection)
- Defense in Depth: layer multiple controls — no single control is sufficient
- Network segmentation: VLANs isolate broadcast domains; routing boundaries enforce policy
- Principle of Least Privilege: devices and users get only the access they need — nothing more



ACCESS CONTROL LISTS: STANDARD VS EXTENDED

- **Standard (1–99): filter by source IP only**
- **Extended (100–199): filter by src+dst IP, protocol, and port**
- **Both evaluated top-down; implicit deny any at the end**
- **Named ACLs: more readable, easier to edit**
- **Standard applied close to destination**
- **Extended applied close to source**



Extended ACL example:

```
ip access-list extended BLOCK_TELNET  
  
deny tcp 192.168.20.0 0.0.0.255 any eq 23  
  
permit ip any any
```

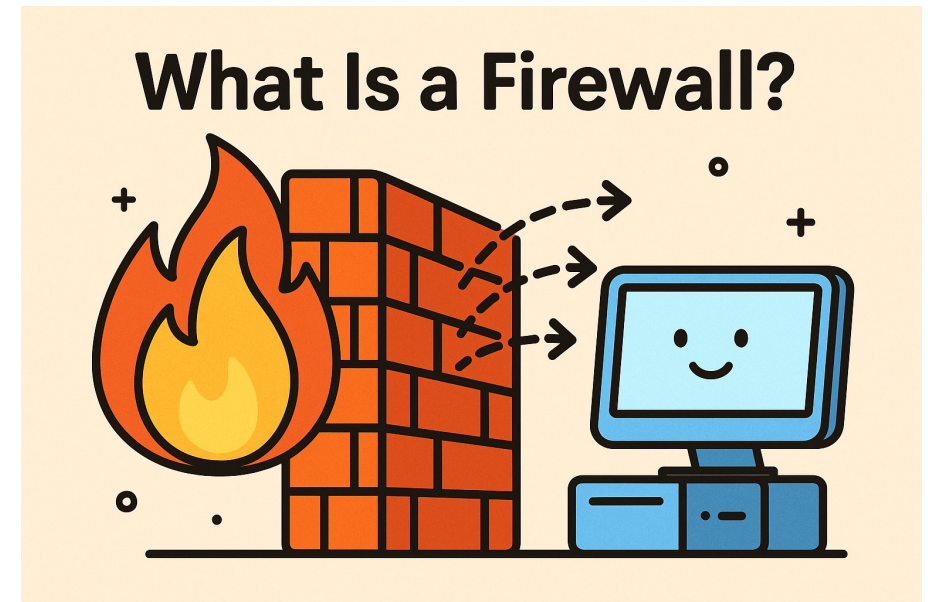
Apply to interface:

```
interface Gi0/1  
  
ip access-group BLOCK_TELNET in
```

Verify: show access-lists

FIREWALLS & PORT SECURITY

- **Packet filter (stateless):** simple rules, no connection tracking
- **Stateful firewall:** tracks connection state — only permits return traffic for established sessions
- **NGFW (Next-Gen):** deep packet inspection, application awareness, IPS integration
- **DMZ:** public-facing servers placed here, isolated from internal LAN
- **Port Security (switch):** limits MACs per port — prevents MAC flooding attacks
- **Config:** switchport port-security maximum 2 → switchport port-security violation shutdown
- **Sticky MAC:** dynamically learns and stores first connected MAC (persist after reload)



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Which type of NAT allows thousands of devices to share a single public IP using port numbers?

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POP QUIZ:

Where should extended ACLs be applied relative to the traffic source?

- A. Far from the source, near the destination
- B. As close to the source as possible
- C. Only on WAN-facing interfaces
- D. On every switch access port



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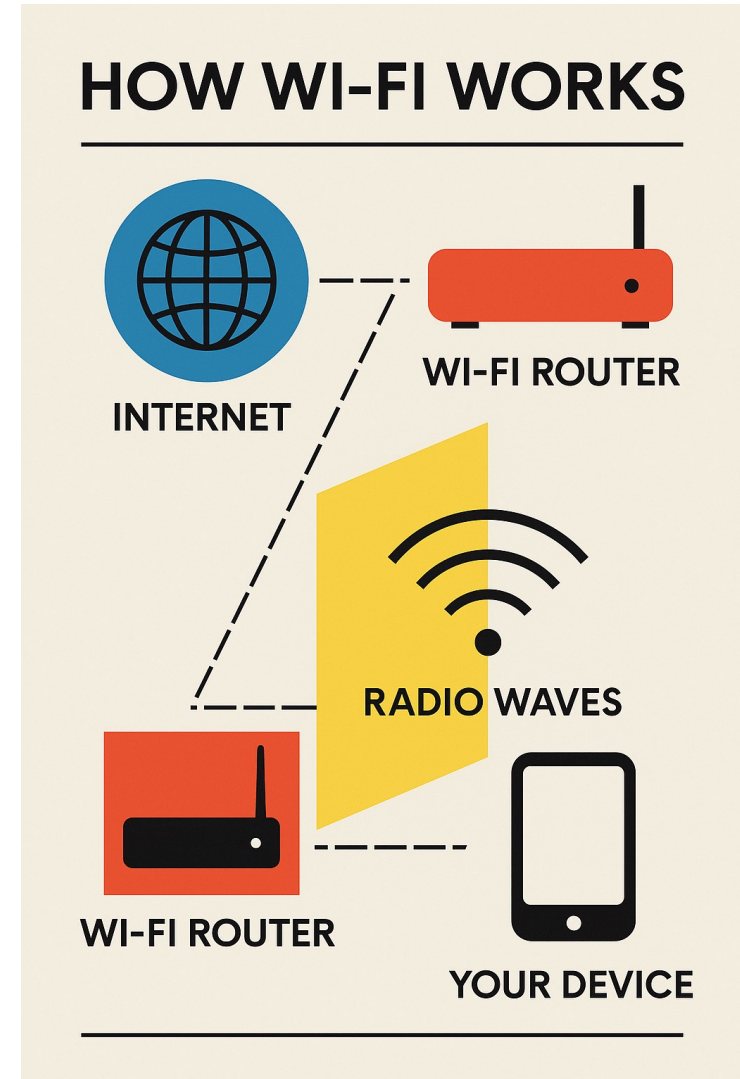
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WIRELESS & VPN FUNDAMENTALS

WI-FI FUNDAMENTALS

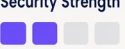
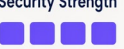
- **IEEE 802.11 standard family** — transmits over radio frequencies
- **2.4 GHz:** longer range/wall penetration, only 3 non-overlapping channels (1,6,11), congested
- **5 GHz:** faster speeds, 24+ non-overlapping channels, shorter range, less interference
- **Wi-Fi 6 (802.11ax):** also supports 6 GHz; improves performance in dense environments
- **SSID:** network name; **BSS** = one AP + clients; **ESS** = multiple APs with same SSID (roaming)
- **Access Points** connect wirelessly to the wired LAN — clients get same services as wired hosts



WIRELESS SECURITY STANDARDS

▪ Security Evolution

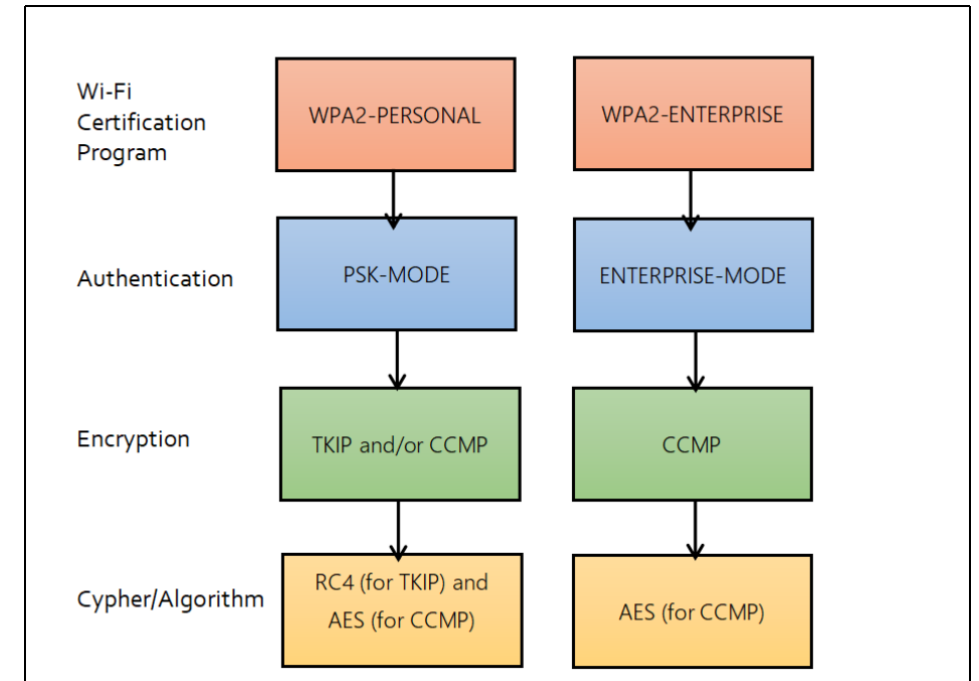
- **WEP (1997): completely broken — never use**
- **WPA (2003): TKIP encryption — RC4 flaws, avoid**
- **WPA2 (2004): AES-CCMP — still secure with strong passwords**
- **WPA3 (2018): SAE replaces PSK — resistant to offline dictionary attacks**
- **Wi-Fi 6 devices require WPA3**
- **Always choose WPA2 minimum; prefer WPA3; disable WEP/WPA**

WEP	WPA	WPA2	WPA3
			
 Year Introduced 1997	 Year Introduced 2003	 Year Introduced 2004	 Year Introduced 2018
 Key Encryption Method RC4	 Key Encryption Method TKIP	 Key Encryption Method AES	 Key Encryption Method SAE (Dragonfly) + AES
 Security Strength Weakest	 Security Strength Weak	 Security Strength Strong	 Security Strength Strongest
			

WIRELESS SECURITY STANDARDS

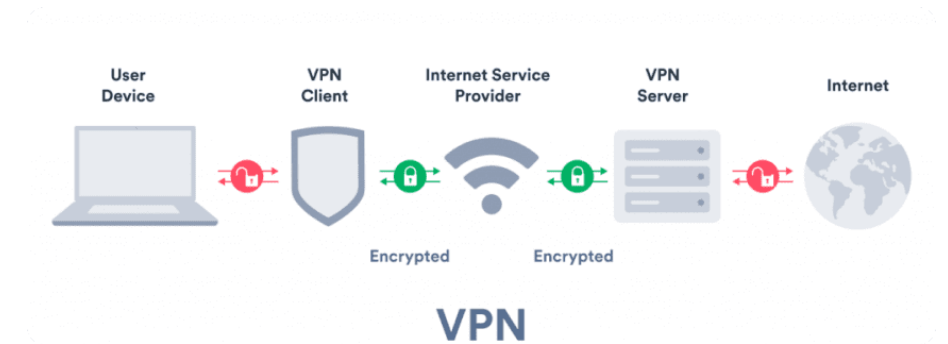
▪ Enterprise vs Personal

- **Personal (PSK):** shared password for all clients — simple, OK for home/small office
- **Enterprise (802.1X):** per-user authentication via RADIUS server — no shared password
- **802.1X flow:** client → AP (authenticator) → RADIUS server → permit/deny
- **Dynamic VLAN assignment possible with 802.1X** — user lands in their VLAN automatically
- **Certificate-based 802.1X:** strongest — no password to steal
- **WPA3 Enterprise:** 192-bit security mode for government/financial



VPN BASICS

- **VPN creates an encrypted tunnel over an untrusted network (Internet)**
- **Site-to-Site VPN: permanent tunnel between two locations — router-to-router**
- **Remote Access VPN: individual users connect securely to corporate LAN from anywhere**
- **IPsec: most common protocol suite — IKE for key exchange, ESP for encryption**
- **SSL/TLS VPN: browser-based or thin client — easier for remote users**
- **Split tunneling: only corporate traffic through VPN; personal traffic goes direct**
- **Full tunnel: ALL traffic routed through VPN — maximum security, higher latency**



POP QUIZ:

Which Wi-Fi security standard uses AES-CCMP encryption and is widely considered secure today?

- A. WEP
- B. WPA
- C. WPA2
- D. TKIP



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LAB 04: DHCP, DNS, NAT LAB

- Goal: Configure all three services on a single router acting as gateway
 - Build: Router (gateway) — Switch — 3 PCs; router WAN port to Internet cloud
 - Configure DHCP pool: 192.168.1.0/24, gateway .1, DNS 8.8.8.8, exclude .1–.5
 - Verify all 3 PCs receive IPs via DHCP: show ip dhcp binding
 - Configure a DNS server in PT: add A records for 'server.local' → 10.0.0.100
 - Test name resolution from PCs: ping server.local should resolve and reply
 - Configure PAT: mark inside/outside, ACL for 192.168.1.0/24, overload on WAN interface
 - Verify NAT: ping simulated Internet from PC; show ip nat translations shows mappings

LAB 05: ACL & WIRELESS SECURITY LAB

- Goal: Apply ACLs to control traffic and configure WPA2 on a wireless AP
 - Build: Router with 2 LAN subnets (192.168.10.0/24 and 192.168.20.0/24)
 - Verify inter-subnet connectivity works before applying ACLs
 - Write extended ACL: deny Telnet (TCP 23) from 192.168.20.0 to 192.168.10.0; permit all else
 - Apply ACL inbound on the 192.168.20.0 interface; test: Telnet should fail, ping should work
 - Add a Wireless Router to the topology — connect to the LAN
 - Configure SSID 'LabNet' with WPA2-PSK; test wireless laptop connectivity
 - Change to WEP and observe the security warning — explain why WEP is unsafe; revert to WPA2

LAB 06: COMPREHENSIVE TROUBLESHOOTING SCENARIO

- **Goal:** Diagnose and fix a broken network using structured troubleshooting methodology
 - **Build:** 2 routers, 2 switches, 4 PCs. These are two sites connected via serial link
 - **Identify:** Run baseline pings between all PCs and record what works and what doesn't
 - **Fault 1:** A PC cannot reach remote hosts. Investigate its network settings
 - **Fault 2:** Ping fails in one direction despite correct routing — check what's filtering traffic on the router
 - **Fault 3:** Switch management IP is unreachable remotely — check its Layer 3 config
 - **Fault 4:** PCs get an IP via DHCP but can't resolve domain names — check the router's DHCP scope
 - **Fix & Document:** Change one thing at a time — record the root cause and OSI layer for each fault